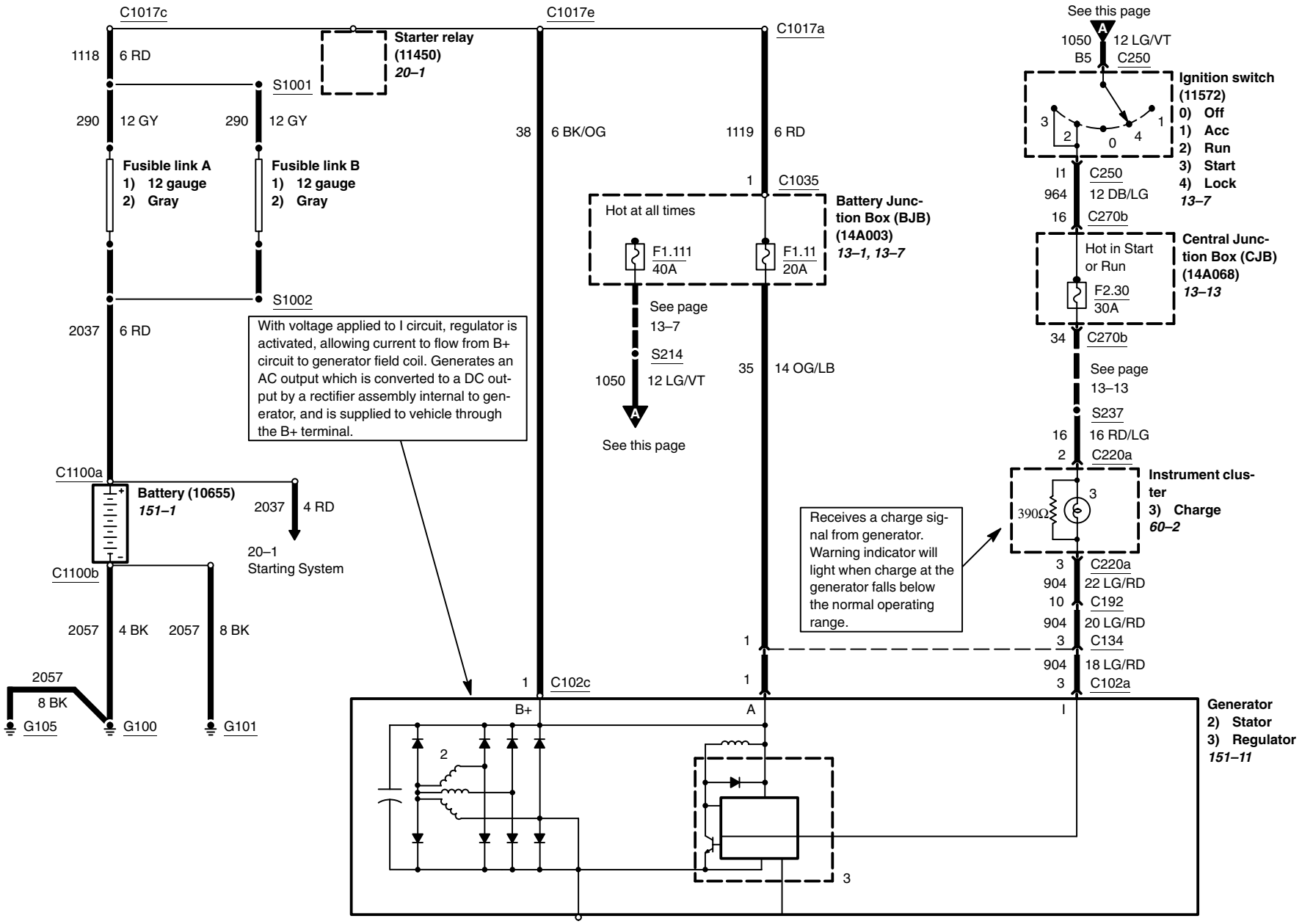4G generator